

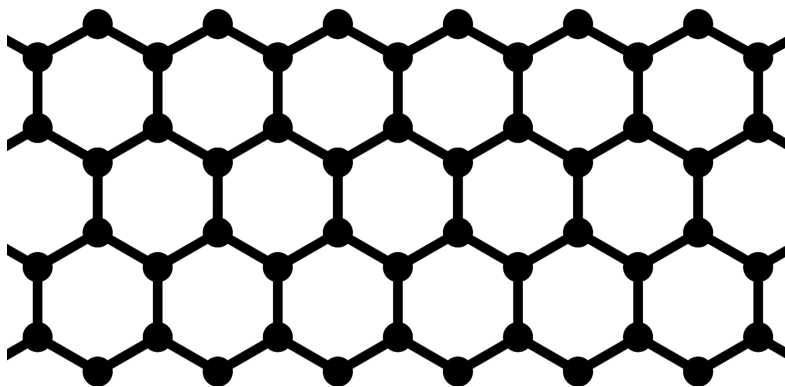
Targeted Marketing: E-Commerce Customer Segmentation via K-Means Clustering

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September 30, 2026

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 Project Repository



Contents

1	Executive Summary	2
2	Introduction	3
3	Methodology	3
4	Results	5
4.1	Deriving the Optimal Number of Clusters, k	5
4.1.1	Elbow Method	5
4.1.2	Silhouette Score Method	5
4.1.3	Hierarchal Clustering Dendrogram Method	6
4.2	K-means Clustering	8
5	Conclusion	10

1 Executive Summary

Businesses that sell online collect vast amounts of data about their customers, but turning that into practical marketing decisions remains a challenge with a lucrative solution. Understanding who customers are and how they shop is central to improving revenue, retention, conversion, and loyalty. This report addresses a specific problem: how can a large, cluttered record of around one million product orders be distilled into meaningful and explainable customer groups that a business can act upon?

An automated analysis pipeline is built to clean an international e-commerce dataset, removing duplicates and condensing a customer's history into five simple traits: how often they order, how recently they last ordered, how much revenue they generate, their age, and the typical price of items they buy.

A machine learning technique grouped customers into five customer groups, yet this begs the question: why five groups? This value was systematically derived through three independent approaches which all confirmed that five was the optimal splitting, consistent with subsequent visualisations which showed they were largely distinct. The characteristics of customers within each group can be interrogated to inform customer personas, allowing for targeted product development and marketing.

The five customer types were: lapsed customers, highly active top spenders, older customers, younger customers, and infrequent buyers of expensive items. Each comes with tailored recommendations, such as win-back offers for lapsed customers and social media campaigns for younger ones. The next step is to test these strategies through controlled and statistically provable experiments to confirm that targeted marketing measurably increases sales.

2 Introduction

Organisations generate significant volumes of customer data which contains useful information to help form a better understanding of the customer base. Insights can be derived for targeted business efforts to improve KPIs such as revenue, conversion, retention, loyalty, satisfaction, and reviews.

Customer segmentation via clustering algorithms help to profile customer types so products can be developed and marketed in a way which maximises revenue. Properties of these clusters can be interpreted to understand who the customers are, how they shop, and how to make it easier for them to make more purchases.

3 Methodology

An e-commerce customer order dataset of 951669 records is used [1]. One record corresponds to one product that is placed within an order by a customer. One customer can make multiple orders, and an order can contain multiple products.

This dataset contains NaN values in some columns, yet these are not removed since they occur in features which are not used in subsequent feature engineering, i.e. the remainder of the respective record contains useful information which should be included. A total of 21 duplicate records are found and removed as to not give these data points any preferential weighting.

Five features are engineered from the data’s original features as input to clustering models, as seen in Table 1. The dataset is then aggregated on Customer ID such that each record corresponds to a different customer, with key summary characteristics stored within the new features.

Feature	Description
Frequency	Number of orders made by a customer
Recency	Days since a customer’s last order
Customer Lifetime Value (CLV)	Total revenue generated by a customer
Customer Age	Approximate Customer Age
Average Unit Cost	Average value of products ordered by a customer

Table 1: Customer Feature Definitions

Once aggregated, exploratory data analysis is conducted to better understand the data. As expected, CLV shows a strong positive correlation with Frequency, confirming that a customer that orders frequently is more likely to have a higher CLV, or vice versa since causality is not defined in these correlations. Moderate negative correlations are identified between Recency and both Frequency and CLV, validating the aforementioned positive correlation since a longer time since a customer’s most recent order means a shorter duration to increase order frequency (across the time span of the dataset), hence reducing overall CLV.

Feature-wise box plots are visualised in Figure 1, with the overall process workflow seen in Figure 2.

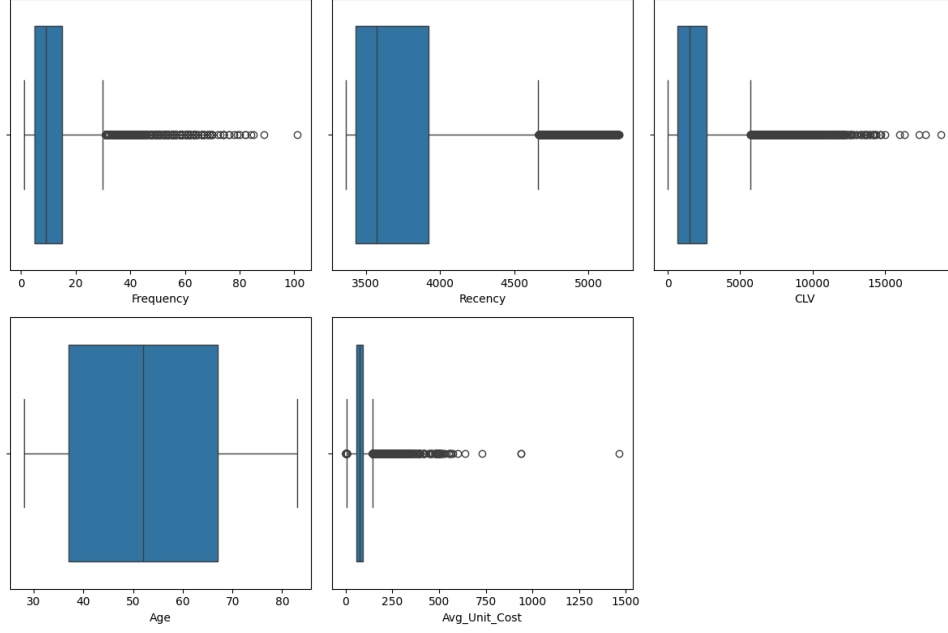


Figure 1: Feature-wise box plots for the engineered features. Significant outliers identified for all features bar Age. Outliers are not few nor far between so likely represent real customers with useful profiles which should be kept for clustering purposes as they may represent a small yet significant customer demographic/type.

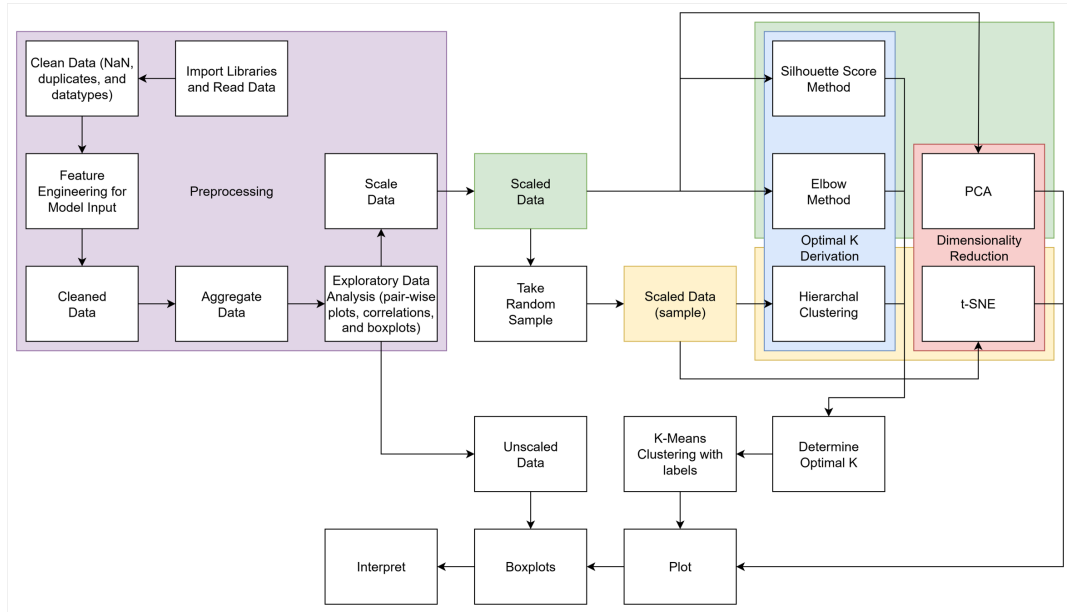


Figure 2: Visualisation of methodology. Preprocessing prepares data for subsequent methods (purple box). Scaled data is used for Silhouette Scores, Elbow Method, and PCA (green box), whereas a sample of the scaled data is used in Hierarchal Clustering and t-SNE due to computational expense (yellow box). PCA and t-SNE are dimensionality reduction techniques (red box), and all other methods are used to derive the optimal number of clusters for k-means clustering (blue box). K-means clustering is then performed and visualised on PCA and t-SNE axes. Feature/cluster-wise boxplots are generated for interpretation to understand cluster customer profiles.

4 Results

4.1 Deriving the Optimal Number of Clusters, k

Three methods are used to derive the optimal number of clusters for customer segmentation with this dataset, each seen in 4.1.1, 4.1.2, and 4.1.3.

4.1.1 Elbow Method

The Elbow Method runs k-means clustering on the dataset across a range of k to calculate the the Within-Cluster Sum of Squares (WCSS), equivalent to the loss function, for each. This is plotted against k to work out which k minimises the WCSS at minimal complexity, the point where the plot starts to decrease linearly, as seen in Figure 3.

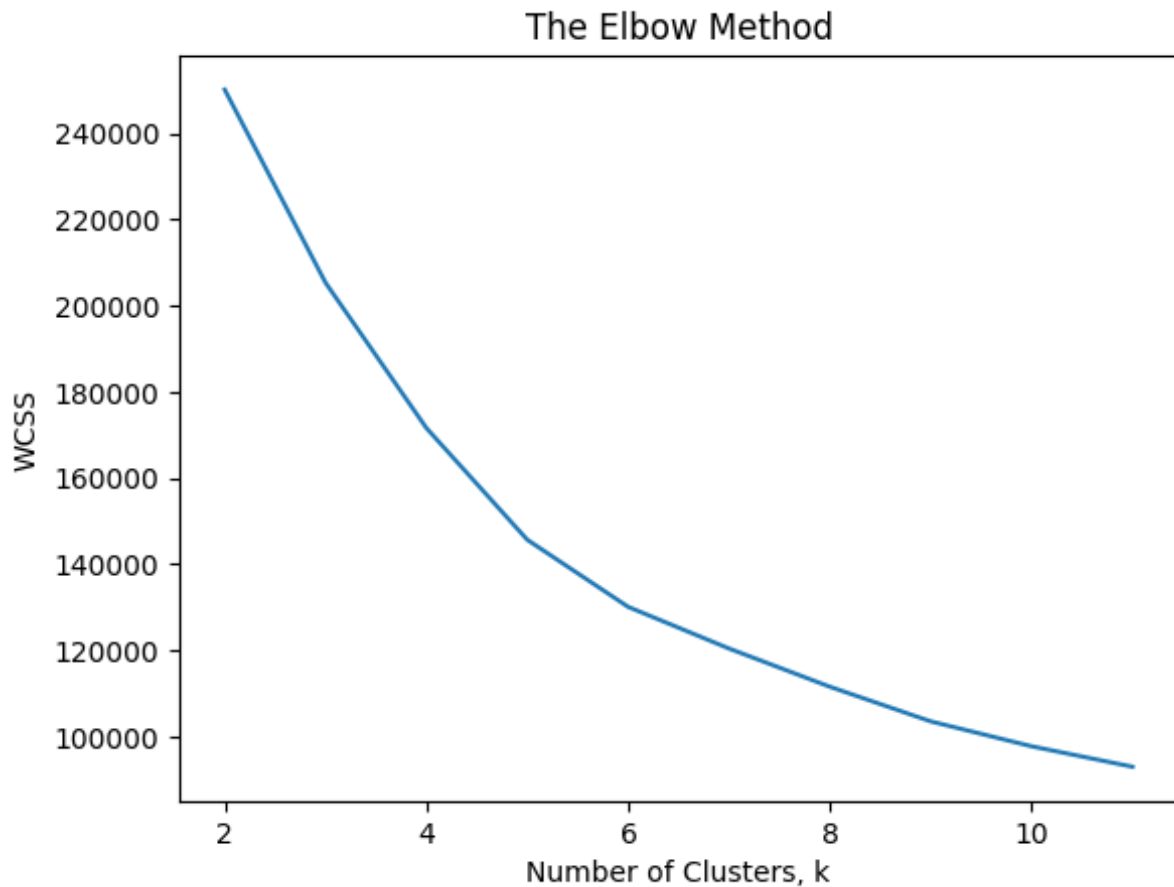


Figure 3: WCSS plotted against k to derive optimal number of clusters for the dataset. No clean "elbow" is seen as the plot gradually approaches a linear decline, so no single value is obviously the "best". Therefore, a range of $k = 5-6$ is derived. Before this range, the plot is descending non-linearly, whereas after the plot descends linearly.

4.1.2 Silhouette Score Method

Silhouette scores compare the average intra/inter-cluster distances across a dataset. A similar method is employed to the Elbow Method: Silhouette Scores are derived and

compared over a range of k , as seen in Figure 4. The optimal number of clusters is the one which maximises the Silhouette Score, which spans the range $[-1, 1]$.



Figure 4: Silhouette Score plotted against k . The maximum value of this plot corresponds to $k = 5$, serving as the optimal number of clusters for this method. Beyond $k=5$, a sharp decline of Silhouette Score is seen, signifying that increasing the number of clusters beyond five leads to more soft clustering. However, it should be noted that when this selection of k is plotted against the function’s range, it appears mostly flat. Computation across the functions entire domain, all k , may show the decline continues, yet this is computationally too expensive to visualise here.

4.1.3 Hierarchal Clustering Dendrogram Method

Hierarchal clustering methods build a hierarchy either from the bottom-up (agglomerative), initialising one cluster per data point and then merging based on Euclidian distance in the feature space, or top-down (divisive), starting with one cluster containing all data and then continually splitting into smaller clusters. These methods afford a dendrogram, a visualisation of clustering with respect to distance, which can be interpreted with intuition to derive a sensible number of clusters.

Agglomerative, bottom-up, clustering is performed on a sample of the dataset due to significant computational expense, with its dendrogram visualised in Figure 5.

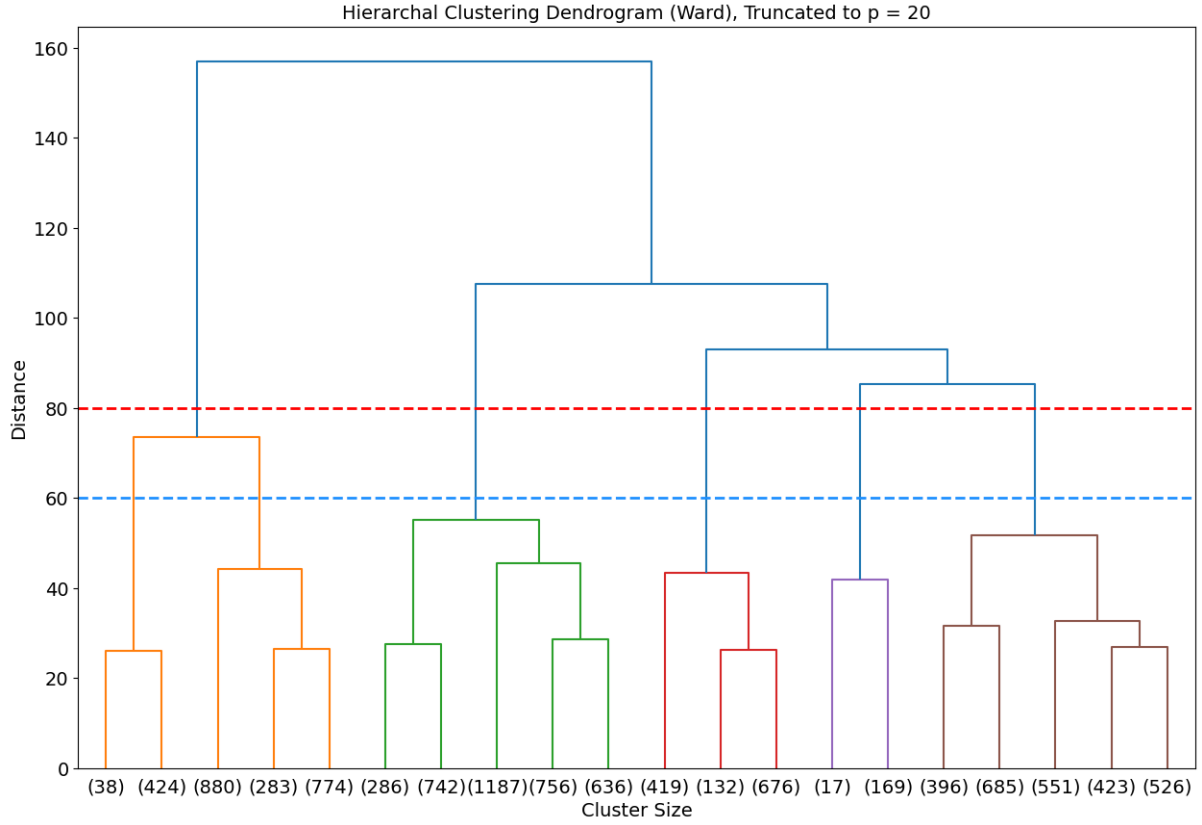


Figure 5: Dendrogram for agglomerative hierarchal clustering performed with Ward linkage, truncated to the final 20 clusters, each containing the number of data points shown on the x-axis. The number of clusters at any given distance is found by how many lines are intersected by its horizontal cut. An example of this is seen by a horizontal dashed red line at $y = 80$, splitting into five clusters, or the blue line at $y = 60$ splitting into six. Optimal k can be found by rough intuition, looking for a horizontal cut which splits large distance gaps to the next cluster merge. From this dendrogram, two k values seem feasible: $k = 5$ and $k = 6$, corresponding to cuts visualised by the red and blue lines, respectively. These retain quick cluster merging below well, with larger merging distances seen above.

4.2 K-means Clustering

The aforementioned methods in Sections 4.1.1, 4.1.2, and 4.1.3 each give an estimate to the ideal number of clusters for this dataset, as summarised in Table 2.

Method	Optimal k
Elbow	5–6
Silhouette Score	5
Hierarchcal Clustering Dendrogram	5–6

Table 2: Optimal Number of Clusters from Different Methods

Overall, $k = 5$ is chosen as the ideal number of clusters for this data since it appears in all three methods. As such, K-Means clustering is done on the data with $k = 5$. This generates five cluster labels of 0–4, inclusive, which are visualised on the PCA and t-SNE axes in Figure 6.

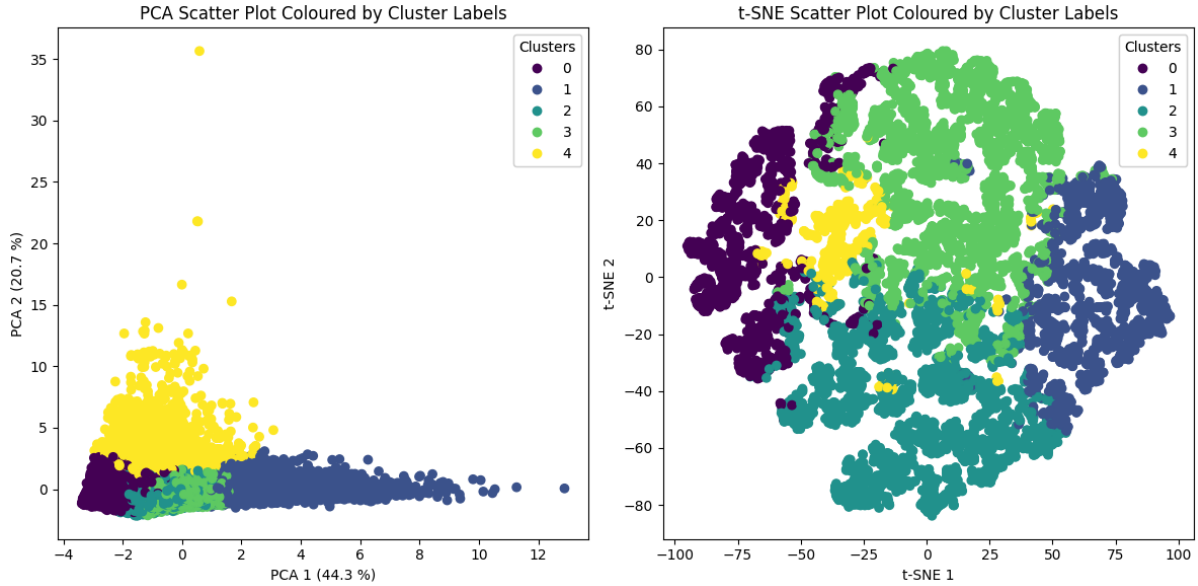


Figure 6: Visualisation of data on PCA (left) and t-SNE (right) axes. Both show areas where one cluster dominates. PCA visualisation shows softer clustering between clusters 2 and 3, whereas t-SNE visually separates these clusters well. Otherwise, clusters 0, 1, and 4 show hard clustering in PCA and similar is seen with t-SNE with the exception of cluster 4 which shows some of its data points further from the main cluster and intersecting with other cluster types.

To gain further insight into the profile of each cluster, box plots for each feature with clusters separated are visualised in Figure 7 to interpret how features vary across clusters.

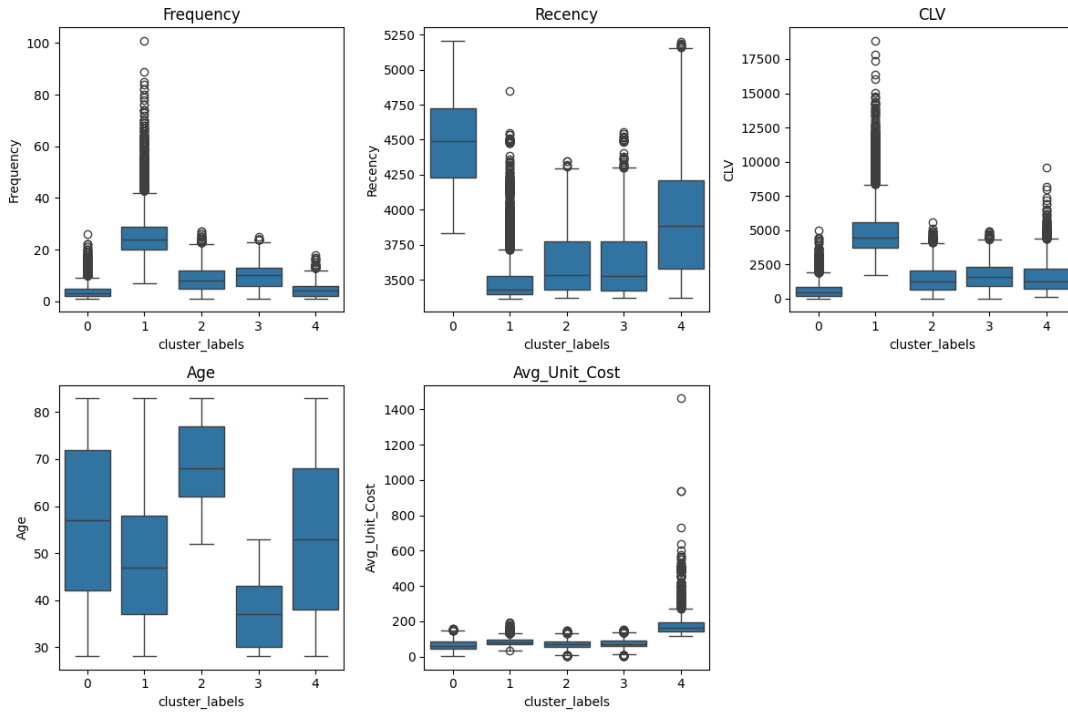


Figure 7: Boxplots of the engineered features with all clusters’ values shown. Interpretation of these boxplots affords proposed customer profile information that is summarised in Table 3.

Cluster Label	Customer Profile Description	Business Suggestion	Further Comment
0	Inactive customers that have not ordered recently, hence lower CLV.	Marketing efforts or return offers could be used to regain custom and make them revenue generating again.	Further product and review analysis to reveal underlying issues (e.g. QC) to improve retention.
1	Active customers who frequently reorder, providing the highest CLV.	Increased marketing efforts and "flash" sales may work due to existing brand trust reducing purchase friction.	Confirms the strong correlations observed between Frequency, Recency, and CLV.
2	Older customer base.	May respond better to postal or phone-based marketing.	Shows similar values to other clusters except for Age.
3	Younger customer base.	Likely prioritise convenience and trendy marketing through social media or email; comedic or catchy branding combined with discounts (e.g. student).	Inverse of Cluster 2. Dataset spans 2012–2016, so ages are approximately 10 years higher than present, though conclusions remain valid when adjusted.
4	High unit value but relatively infrequent customers.	Targeted marketing for accessories/smaller items to accompany higher-priced purchases.	Higher order values offset low purchase frequency, resulting in comparable CLV to other clusters.

Table 3: Summary of Customer Clusters with Business Interpretations

5 Conclusion

Customer product and order data is highly dimensional, making it difficult to uncover any meaningful insights. Five features are engineered to collapse the dimensionality whilst retaining strong customer profile character in the remaining features. These can be clustered via the k-means algorithm, however, this requires the number of clusters to be specified on input.

Therefore, three methods are used to derive the optimal number of meaningful clusters: Elbow Method, Silhouette Scores, and Hierarchical Clustering Dendrogram. These methods arrive on a value of $k = 5$, segmenting customers into one of five clusters.

These are visualised on dimensionally reduced PCA and t-SNE axes, both showing areas of the plots reserved for each cluster type. PCA shows some overlap between clusters 2 and 3, whereas t-SNE generally separates clusters better but does not preserve as much explainability since it is not a linear dimensionality reduction method.

Customer segmentation via K-means clustering provides useful business information as to who a business' customers are, allowing for recommendations as to how to increase future revenue from each cluster. This is done by making assumptions about what a certain customer profile might respond well to. This can be tested by a statistical method, such as a paired t-test, to find if there is any statistically significant difference between two random samples of the cluster that do/do not have targeted efforts applied.

References

- [1] SAS. Customers clean (revised 15/12/2021). 2024. Accessed 07/04/2026.